

Jacobian Free Newton Krylov Approach for Navier Stokes Equation using Nodal Integral Method



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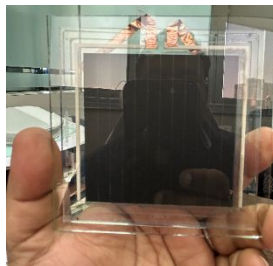
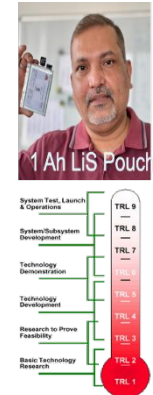
Vehicle to Grid



Solar Thermal



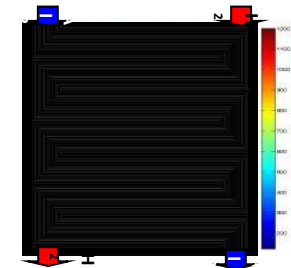
Batteries



Solar Cells



Hydrogen Storage



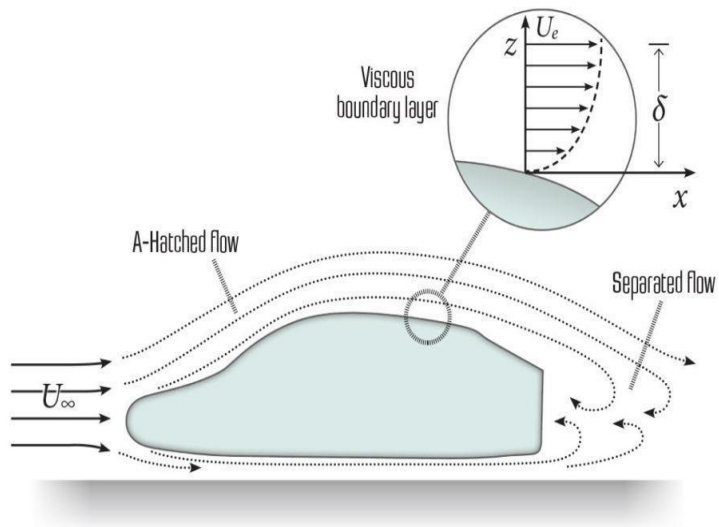
Fuel Cells

Biomass Gasification, Conventional Energy, Energy Systems modelling, Energy Policy

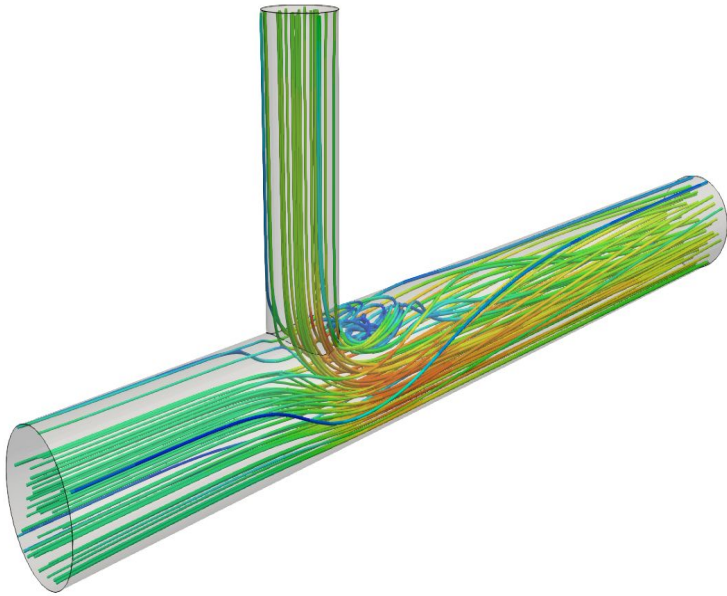
Outline

- ❑ Fluid Flow Problems
- ❑ Computational Complexity of Fluid Flow Problem
- ❑ Newton Method for non-linear equations
- ❑ Krylov Method for a linear set of equations
- ❑ Jacobian Free Newton Krylov
- ❑ Preconditioning
- ❑ Nodal Integral Method
- ❑ Preconditioner for Nodal Methods
- ❑ Results

Fluid Flow Problems



Fluid Flow Problems



Complexity of Fluid Flow

Incompressible Navier-Stokes Equation

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left[\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right] + \rho g_x$$

$$\rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left[\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right] + \rho g_y$$

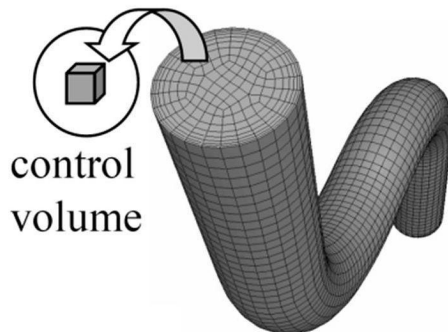
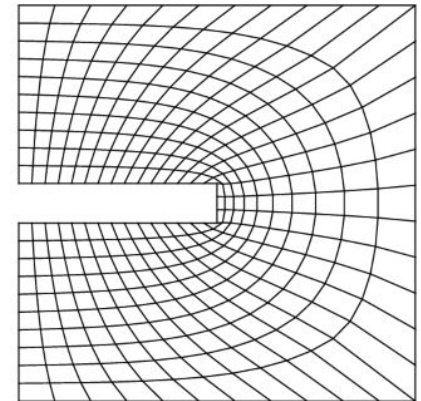
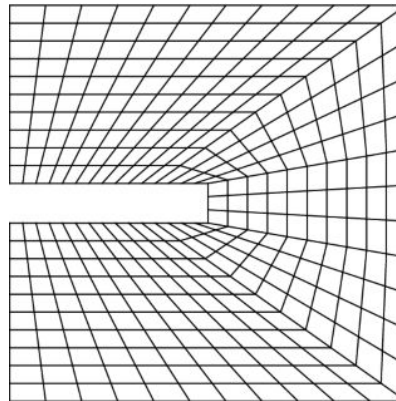
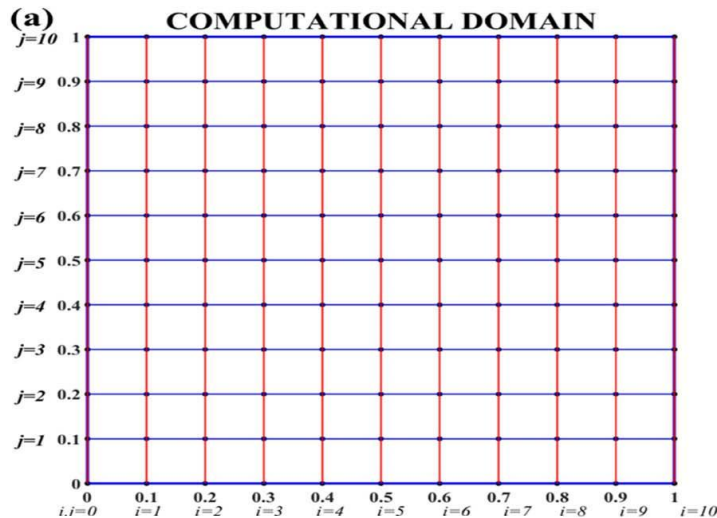
$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho v_x) + \frac{\partial}{\partial y} (\rho v_y) + \frac{\partial}{\partial z} (\rho v_z) = 0$$

- Highly non-linear
- Pressure-velocity coupling

Solving Non-Linear Partial Differential Equations

Domain Discretization



- Simple to complex depending on geometry and flow
- Differential equations converted to difference equations at each point or node
- Number of algebraic equations can range from thousands to millions and beyond

Solving non-linear algebraic equations

- The solution procedure depends on the equations and numerical method
- For linear problems, linear System of algebraic equations $\mathbf{Ax} = \mathbf{b}$. Those millions of equations can be written in matrix form.
- Different matrix structures require different solution methods.
- Non-linear systems require iterative procedures.
- Each iteration of non-linear solver requires solution of a matrix system
- Newton method is one of the approaches for solving such non-linear systems

Newton Method for single variable

Non-linear algebraic equation in the form of $f(x) = 0$

$$x^{n+1} = x^n - \frac{f(x^n)}{f'(x^n)}$$

$f(x) = 0$ is replaced by $f(x^n + \delta) = 0$

Taylor series Expansion and truncation at linear term

$$f(x^n) + \delta f'(x^n) = 0$$

$$\delta = -\frac{f(x^n)}{f'(x^n)}$$

$$x^{n+1} = x^n + \delta$$

Newton Method for multiple variables

Non-linear set of algebraic equation in the form of $F(x) = 0$

Taylor series Expansion and truncation at linear term

$$F(x^n) + J(x^n)\delta(x^n) = 0$$

$$J(x^n)\delta(x^n) = -F(x^n)$$

$$x^{n+1} = x^n + \delta$$

- ❑ **Advantage:** Quadratic convergence (Reduces number of iterations)
- ❑ **Disadvantage:** Create & Store Jacobian matrix at each Newton iteration (Increases Cost per iteration)

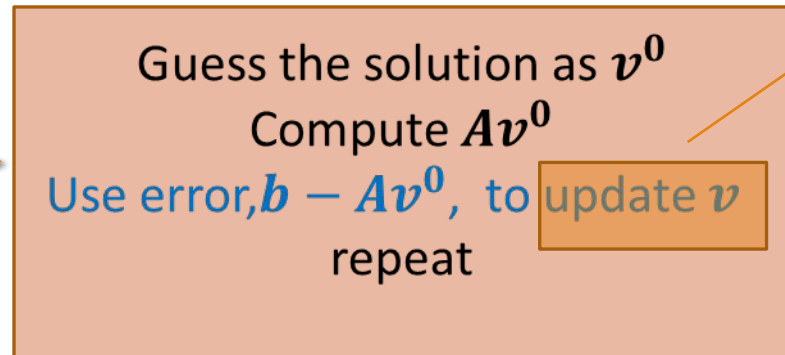
Net cost should be less than Picard Method (linear convergence)

Krylov Subspace methods

$$J(x^n)\delta(x^n) = -F(x^n)$$

is a matrix system $Ay = b$

Krylov



Several approaches,
GMRes used here

Important point is that, we need only **matrix-vector product** and not the matrix.
Our matrix is a Jacobian, so we need **Jacobian-vector product**

$$Jv = \frac{F(x + \epsilon v) - F(x)}{\epsilon}$$

Jacobian-Free Newton-Krylov method (JFNK)

At each Newton iteration

$$J\delta x = -F(x)$$

Advantage: Deals with the matrix-vector product

Jacobian J

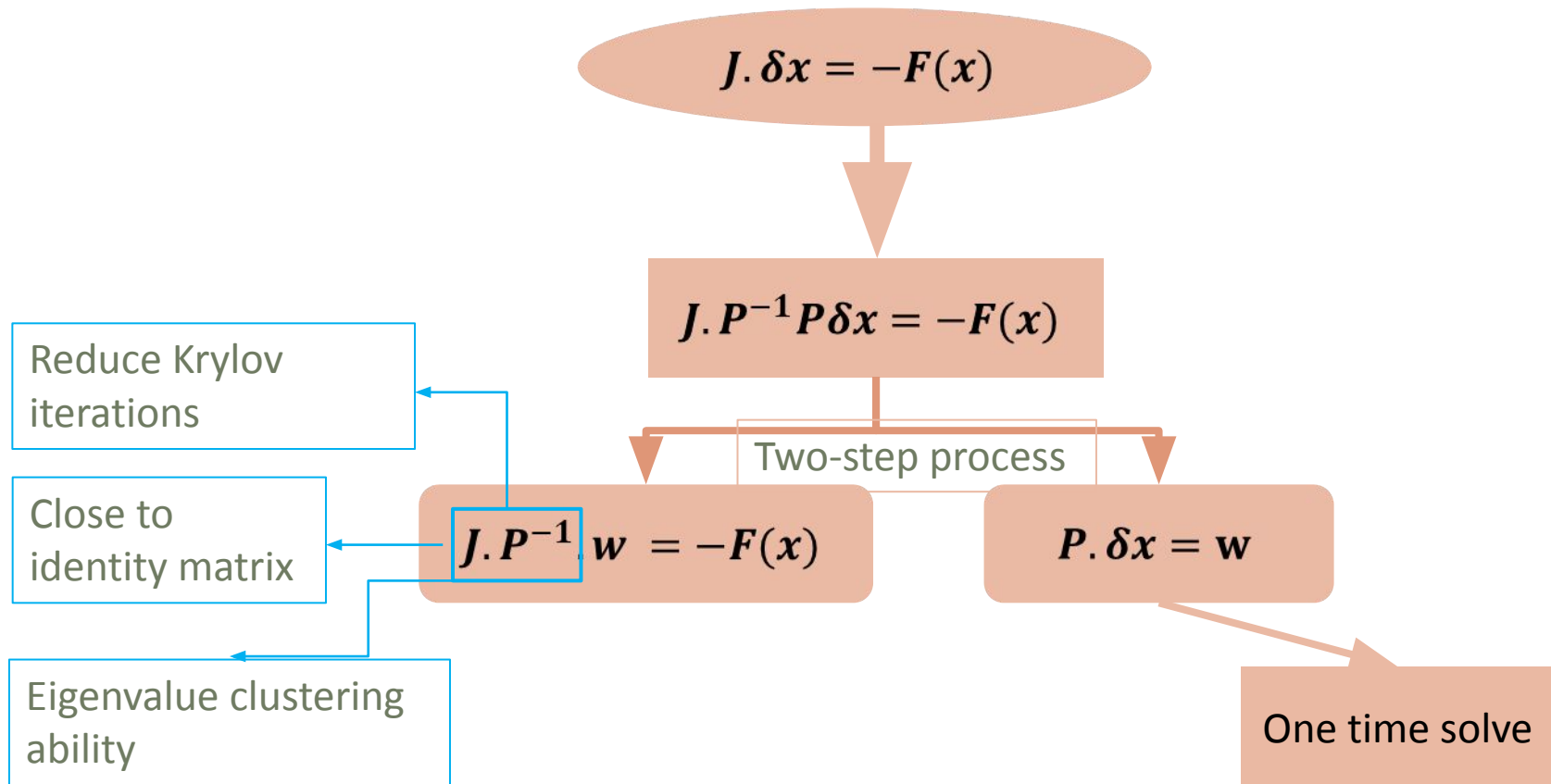
Vector v

$$J \cdot v = \frac{F(x + \epsilon v) - F(x)}{\epsilon}$$

Preconditioning

- Now, we have JFNK method.
- However, the Krylov solvers do not always have good convergence.
- Therefore, preconditioning is needed.
- The design of preconditioners is the main area of research.

Preconditioning



However, this approach should be Matrix (Jacobian) Free

Preconditioned JFNK

First sub-step (i.e., $J \cdot P^{-1} \cdot v \approx (F(x + \epsilon P^{-1}v) - F(x))/\epsilon$) is further split into two sequential sub-steps

First Part

Preconditioning sub-step:
Approximate the solution
of y by solving $Py = v$

Second Part

The matrix-vector product is
approximated by using
 $J \cdot y = \frac{F(x + \epsilon y) - F(x)}{\epsilon}$

The cost of solving the linear system $Py = v$ should be negligible. For this,

1. Not only we need a good P such that $J \cdot P^{-1}$ is close to identity.
2. We also need a good linear solver for the system $Py = v$.

Note that whole crux of the scheme is on the choice/design of the preconditioner.

Preconditioners

- Math based preconditioners

- Multigrid, Incomplete LU factorization etc.

- Physics based preconditioners

- Idea developed for multi-physics problem

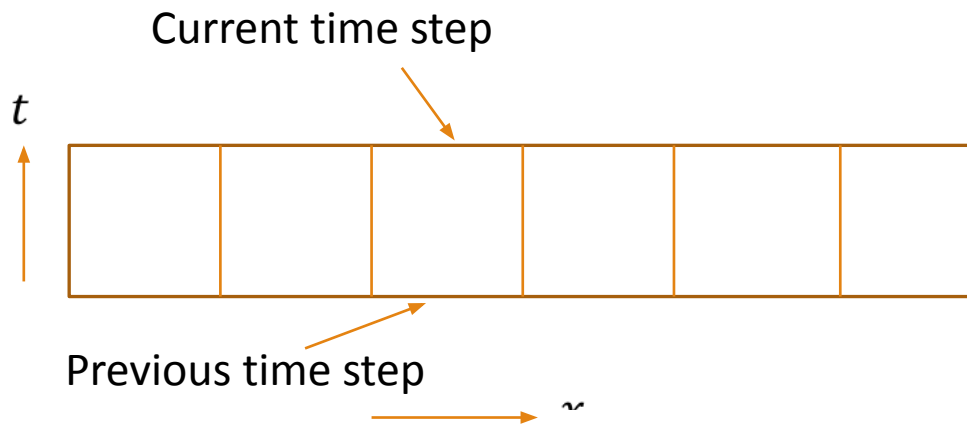
- Use existing efficient solvers for different physics

- Usually heuristic algorithms for weakly non-linear decoupled problems

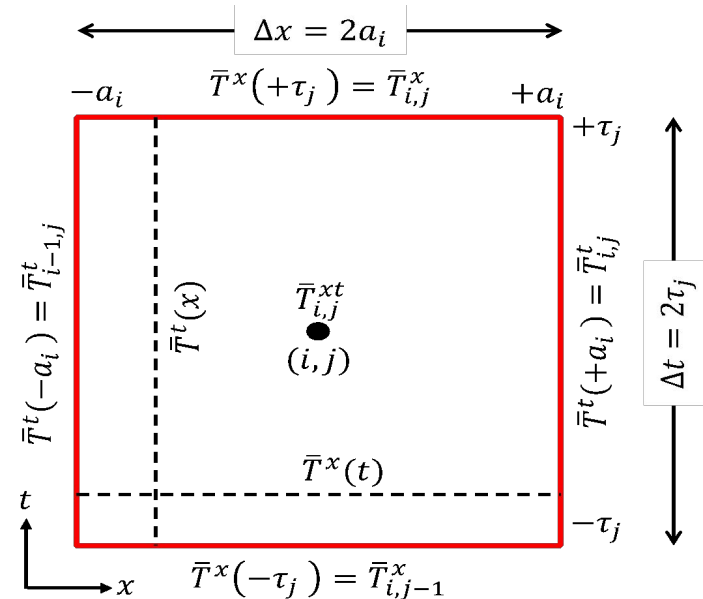
Nodal Integral Method Steps

- Average the governing equation in a node, over x to get an ODE.
- Approximately solve the obtained ODE
- Use continuity at the interface of adjacent nodes to get an algebraic equation
- Repeat the same over t .
- These algebraic sets become interdependent after closure.
- The local semi-analytic solutions capture the physics and lead to higher accuracy.
- The variables are surface averaged quantities.
- For 2D and 3D same process is repeated for all variables.

Nodal Integral Method for Burgers Equation



(A) Domain discretization.



(B) Local coordinate system and transverse-averaged quantities.

Nodal Integral Method

Modified Nodal Integral Method (MNIM)



Modified-Modified Nodal Integral Method (M²NIM)



Consider these equations in a node with u_c and u_p as a kind of node averaged velocity at current and previous time step respectively.

The terms in orange boxes are non-linear terms. Term in blue box is linearized convection. The treatment of these terms in analytical solutions is different.

Final set of algebraic equations for *MNIM*

$$D\bar{u}_{i-1,j}^t + E\bar{u}_{i,j}^t + F\bar{u}_{i+1,j}^t = \left(\frac{\bar{u}_{i,j}^x + \bar{u}_{i,j-1}^x}{2X/Re} \right) G + \left(\frac{\bar{u}_{i+1,j}^x + \bar{u}_{i+1,j-1}^x}{2X/Re} \right) 2aA$$

$$\left(\frac{X - \tau}{\tau} \right) \bar{u}_{i,j}^x - \left(\frac{X + \tau}{\tau} \right) \bar{u}_{i,j-1}^x + 2A\bar{u}_{i,j}^t + 2B\bar{u}_{i-1,j}^t = 0$$

Where all the coefficients contains both linear as well as exponential terms, for example

$$A_{i,j} = \frac{1}{2a_{i,j}u_c Re} - \frac{1}{[e^{2a_{i,j}u_c Re} - 1]} \quad \text{and}$$

$$G_{i,j} = \frac{2a_{i,j}}{[1 - e^{-2a_{i,j}u_c Re}]} - \frac{1}{u_c Re}$$

u_c is defined as average of $\bar{u}_{i,j}^t$ and $\bar{u}_{i-1,j}^t$

The exponential of u_c in coefficients makes the terms highly sensitive leading to ill-conditioned matrices.

Final set of algebraic equations for M^2NIM

$$D\bar{u}_{i-1,j}^t + E\bar{u}_{i,j}^t + F\bar{u}_{i+1,j}^t = \left(\frac{\bar{u}_{i,j}^x + \bar{u}_{i,j-1}^x}{2X/Re} \right) G + \left(\frac{\bar{u}_{i+1,j}^x + \bar{u}_{i+1,j-1}^x}{2X/Re} \right) 2aA + (u_c - u_p) \frac{1}{2a_i} (\bar{u}_{i,j}^t - \bar{u}_{i-1,j}^t)$$

$$\left(\frac{X - \tau}{\tau} \right) \bar{u}_{i,j}^x - \left(\frac{X + \tau}{\tau} \right) \bar{u}_{i,j-1}^x + 2A\bar{u}_{i,j}^t + 2B\bar{u}_{i-1,j}^t = 0$$

Where all the coefficients contains both linear as well as exponential terms, for example

$$A_{i,j} = \frac{1}{2a_{i,j}u_p Re} - \frac{1}{[e^{2a_{i,j}u_p Re} - 1]} \quad \text{and}$$

$$G_{i,j} = \frac{2a_{i,j}}{[1 - e^{-2a_{i,j}u_p Re}]} - \frac{1}{u_p Re}$$

The non-linearity becomes quadratic. For low Re and/or small time step this term is small.

However, when we increase Re and larger time steps, error increases.

Comparison of $MNIM$ and M^2NIM

$MNIM$

Coefficients depends upon u_c and v_c

Exponential nonlinearity

Computationally expensive

Highly accurate solutions

Solver

M^2NIM

Coefficient depends upon u_p and v_p

Quadratic nonlinearity

Computationally inexpensive

Less accurate solutions

Preconditioner after linearization

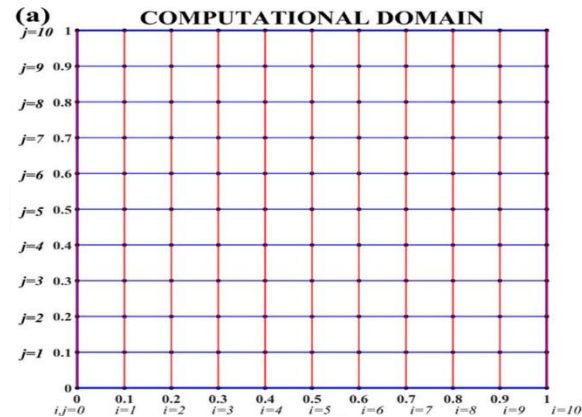
Solver for Preconditioner for Burgers Equation

$$D\bar{u}_{i-1,j}^t + E\bar{u}_{i,j}^t + F\bar{u}_{i+1,j}^t = \left(\frac{\bar{u}_{i,j}^x + \bar{u}_{i,j-1}^x}{2X/Re} \right) G + \left(\frac{\bar{u}_{i+1,j}^x + \bar{u}_{i+1,j-1}^x}{2X/Re} \right) 2aA + (u_c - u_p) \frac{1}{2a_i} (\bar{u}_{i,j}^t - \bar{u}_{i-1,j}^t)$$

$$\left(\frac{X - \tau}{\tau} \right) \bar{u}_{i,j}^x - \left(\frac{X + \tau}{\tau} \right) \bar{u}_{i,j-1}^x + 2A\bar{u}_{i,j}^t + 2B\bar{u}_{i-1,j}^t = 0$$

- We use linearized M^2NIM as preconditioner.
- After eliminating $\bar{u}_{i,j}^x$, the system looks similar to a central finite difference scheme (of course with different coefficient values).
- Therefore, it is a tridiagonal system,
- Solving a tridiagonal system with Thomas algorithm has negligible cost

Solver for Preconditioner for 2D Burgers Equation



- **Alternate Direction Implicit (ADI)**
 - Make initial guess of the solution
 - Solve row-wise tridiagonal systems
 - Solve column wise tridiagonal systems
 - Iterate till convergence
- **Time-Split Alternate Direction Implicit (TS-ADI)**
 - Make initial guess of the solution
 - Solve row-wise tridiagonal systems
 - Solve column wise tridiagonal systems
 - Avoid iteration as it is second order accurate.

Note that Preconditioner requires approximate solution

Solver of Preconditioner for Navier-Stokes Equation

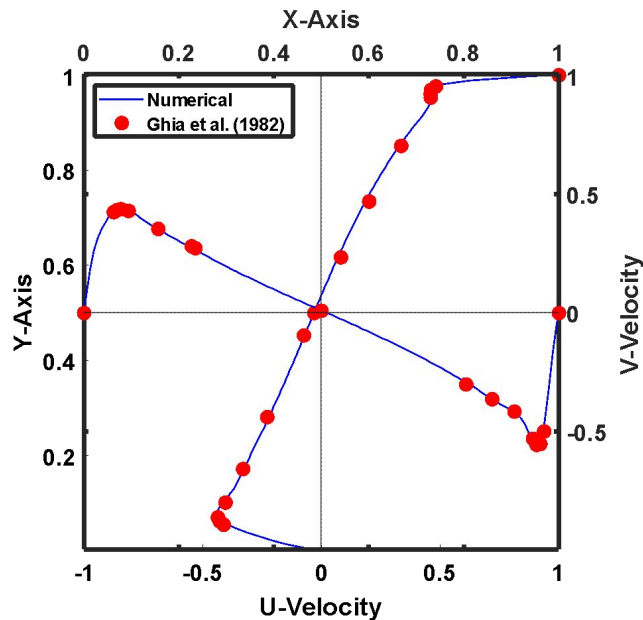
- Momentum equation is similar to Burgers Equation.
- Instead of solving continuity equation, Pressure Poisson Equation is solved.



- The above equation is derived by taking divergence of momentum equation and simplifying using continuity.
- $\mathbf{f}(\mathbf{u})$ has non-linear terms of velocity.
- The coefficient matrix of the LHS is time and iteration independent.
- For preconditioner use operator split (block preconditioner)
- Solve momentum equation by TS-ADI
- Solve pressure equation using Modified Incomplete LU (MILU somewhat standard).

Verification of the developed algorithm

(A) VELOCITY PROFILES WITH GRID
120×120



(B) FLOW STREAMLINES

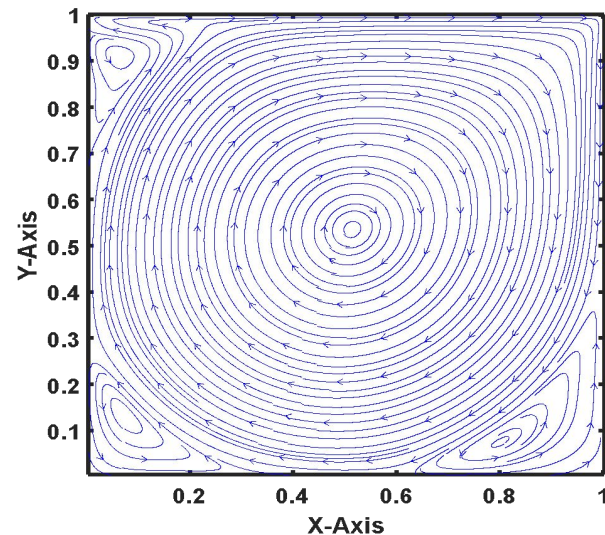
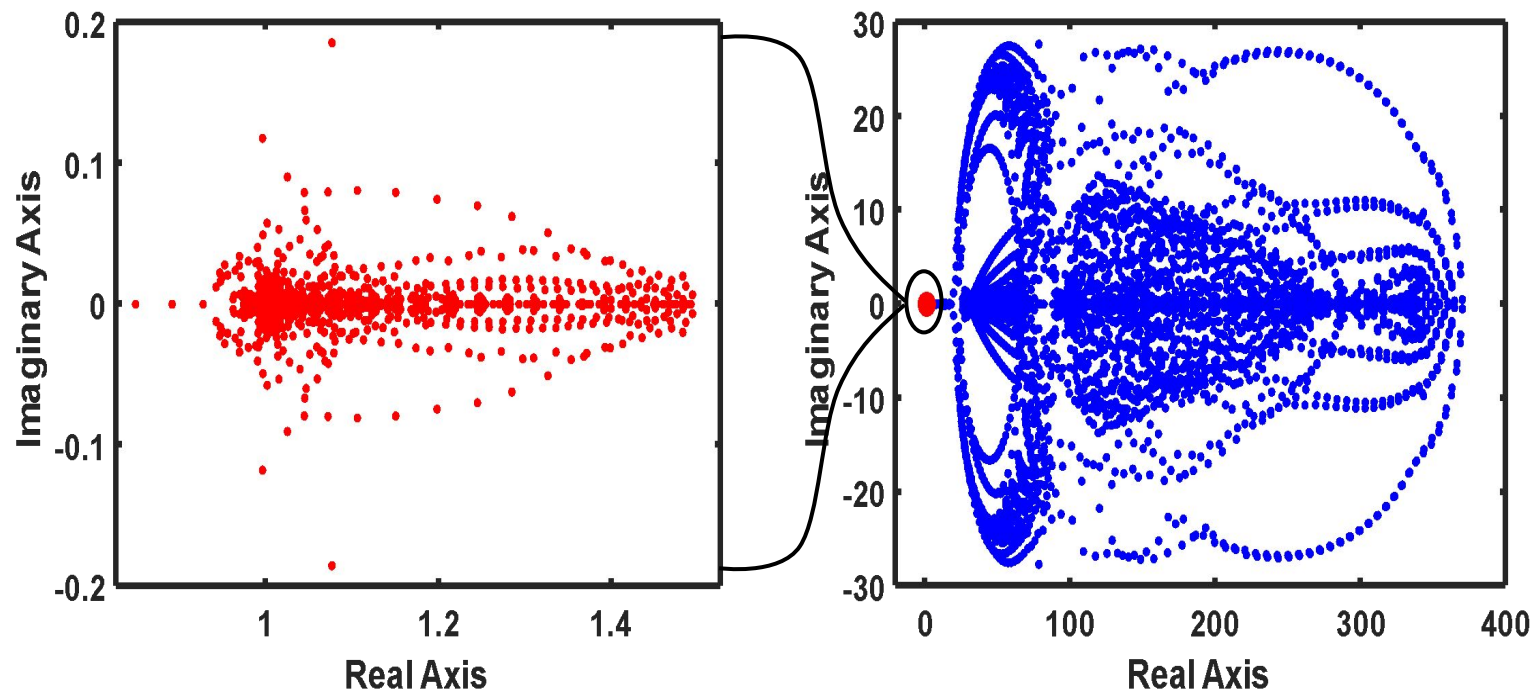


Figure 6: (A) Computed u and v velocity profile along a horizontal and vertical line through the geometric centre of the cavity for $Re = 5000$, solid lines indicate the present simulations with grid size 120×120, and the red circle indicates the simulation of Ghia et al. (1982) [1] with grid 256×256. (B) Flow streamlines at $Re = 5000$.

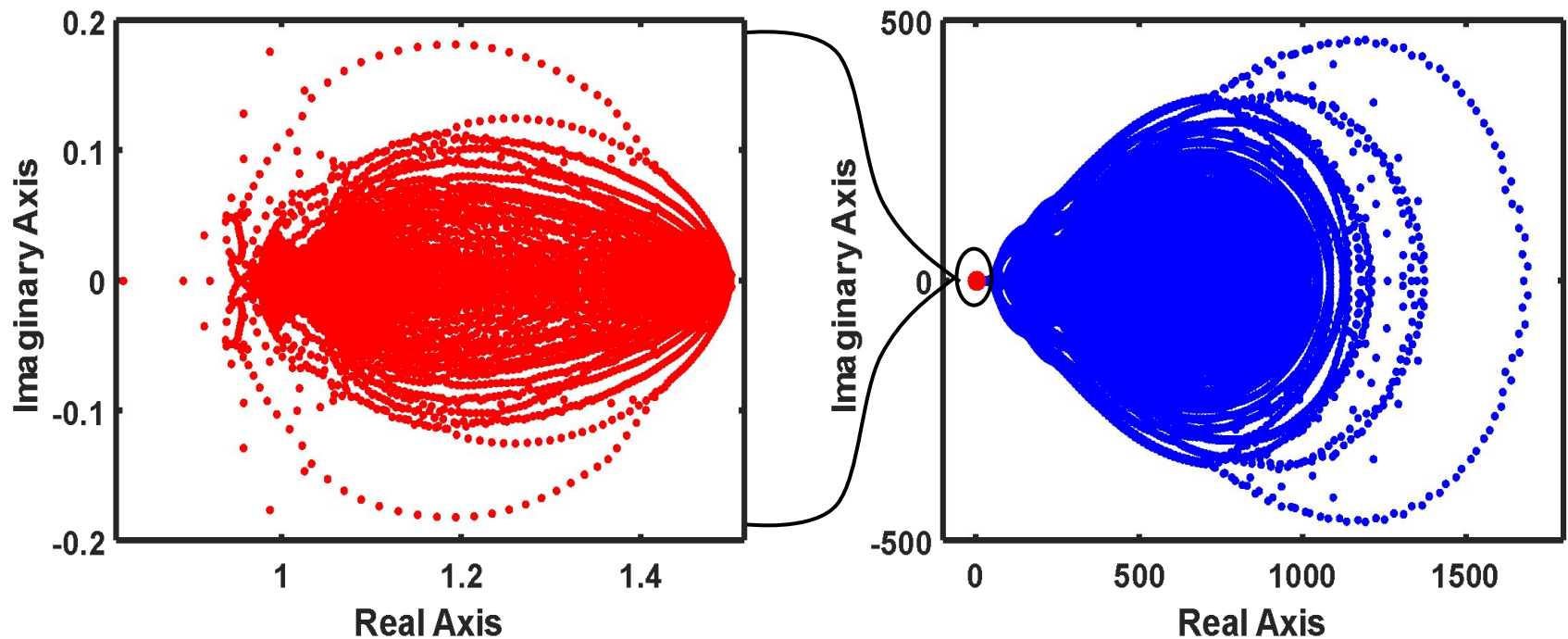
Eigenvalue clustering

Eigenvalues of the J (blue) and JP^{-1} (red) matrix for $\text{Re} = 100$



Eigenvalue clustering

Eigenvalues of the J (blue) and JP^{-1} (red) matrix for $\text{Re} = 1000$



Performance comparison between JFNK vs. PC-JFNK

Comparison of CPU runtime in seconds for JFNK vs. PC-JFNK

	CPU runtime per time step		Time-saving improvement factor		CPU runtime per time step		Time-saving improvement factor
	JFNK	PC-JFNK			JFNK	PC-JFNK	
0.0833	0.34	0.03	11.33	0.0417	6.79	0.11	61.72
0.0417	3.60	0.09	40.00	0.0278	19.24	0.20	96.20
0.0278	10.28	0.19	54.11	0.0208	35.40	0.34	104.12
0.0208	23.39	0.38	61.55	0.0161	65.50	0.55	122.73
0.0167	45.18	0.70	64.54	0.0132	120.34	0.85	141.58

Summary

- JFNK has been successfully applied to NIM based Navier-Stokes solver
- A linearized version of the simplified scheme is used as a preconditioner.
- Unique attributes of the NIM lead to extremely good preconditioning of momentum equation.
- Operator splitting approach is used for pressure-velocity coupling of the preconditioner.
- Significant saving in computational effort are seen for the preconditioned JFNK.

Publications

Nadeem Ahmed, Suneet Singh, Niteen Kumar, “Physics-based preconditioning of Jacobian-free Newton–Krylov solver for Navier–Stokes equations using nodal integral method”. International Journal of Numerical Methods in Fluids, 2023.

Nadeem Ahmed, Suneet Singh, R. P. Bharti, “Modified preconditioned Newton-Krylov approaches for Navier-Stokes equations using nodal integral method”. Computers and Mathematics with Applications, 2025.

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- The contents of this presentation are solely my responsibility and do not necessarily represent the official views of the Fulbright Program, the Government of the United States, or the US India Educational Foundation.

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